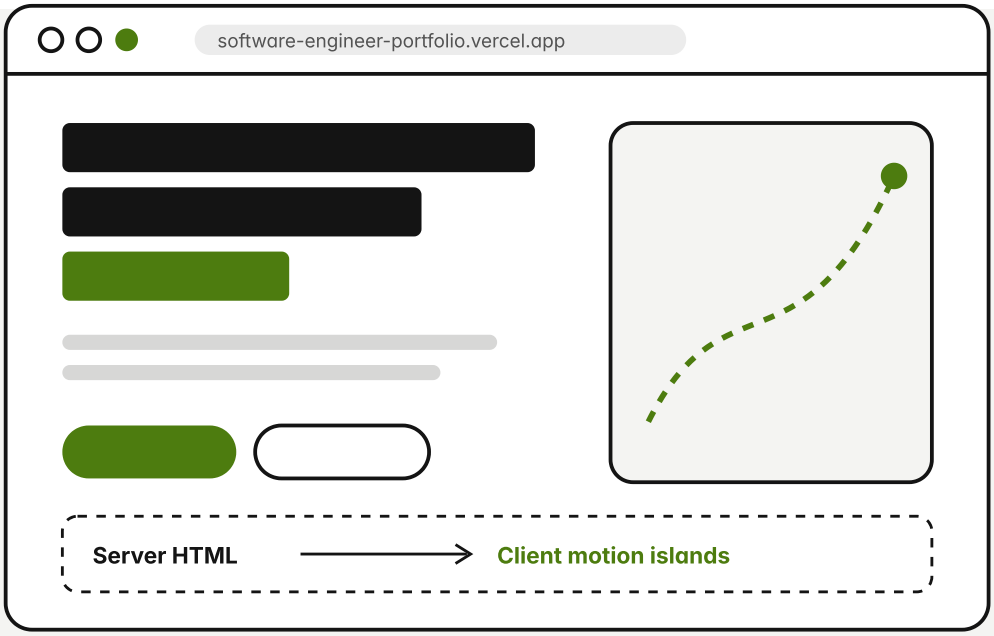
erwin.dev



DESIGN DOCUMENT.

Software Engineer Portfolio

The Webflow portfolio rebuilt in React: motion-heavy, but fast, accessible and leak-free

VERSION	DATE	AUTHOR	STATUS
1.0	8 Oct 2026	Yosaphat Harwindra	Signed off

01 The Problem

Erwin's portfolio exists today as a Webflow site with a GSAP motion layer, live at `blacks-amazing-site-ff73a5.webflow.io`. It has a home page, five case studies (Rover, Fmlx Common UI, ESS, Promptomat, Parish room booking), a downloadable résumé and a motion layer of 28 named modules in `fx.js` (737 lines). It works for applications today, and it stays live (D-02).

For the roles Erwin applies to (remote React and full-stack positions, Europe first), it has three gaps:

1. **No React evidence.** The site is built in the Webflow Designer, so a hiring manager sees design and motion taste but no React code. His strongest work (Rover, Common UI and ESS at Formulatrix) is under NDA and has no public repository, so GitHub alone hides two years of production React.
2. **No ownership.** The free Webflow plan serves the site from a `webflow.io` subdomain. A custom domain needs a paid site plan (Basic, $15 per month billed yearly).
3. **Updates need the Designer.** Every copy change is a Designer session and a publish, with no review step and nothing that checks the change did not break a page.

The project is a React (Next.js) version of the same site, written by Erwin, that is itself the code sample: public, typed, tested and hosted for free.

Why this project (portfolio angle)

A portfolio site usually proves little, because it is static pages. This one carries a heavy motion layer written as plain DOM scripts, and moving it into React has three invisible hard parts:

- **Motion that survives navigation.** The App Router has no "leave" hook. Tweens, ScrollTriggers and SplitText instances leak or duplicate across client navigations unless every effect is scoped and cleaned up. Webflow never had this problem, because every link was a full page load.
- **Heavy motion with a fast first paint.** Content renders as Server Components and only thin client islands ship motion code, so Core Web Vitals stay in the "good" range on mobile.
- **Motion that never hurts access.** Reduced motion is a hard gate, the custom cursor only appears with a fine pointer, keyboard focus stays visible, and every page reads fully with JavaScript off.

The interview story: motion-heavy, but fast, accessible and leak-free, backed by tests rather than screenshots.

Team	Erwin: developer, writes every component and page. Claude Code: mentor for Erwin's code, implementer for chores (asset sync, content types, tests, CI, config, deploys) (D-06)
Planned term	2 weeks target from sign-off, 3 weeks hard cap. Week 1 ends shippable (D-09)
Key technologies	Next.js 16 App Router, React 19, TypeScript, GSAP 3.15 with `@gsap/react`, Lenis, CSS Modules
Full stack	Static site generated at build time from JSON content; no backend, no database. Vitest (content schema), Playwright (Chromium, Firefox, WebKit, axe), Lighthouse CI. GitHub Actions for CI, Vercel Hobby for hosting (`vercel.app` address, custom domain later). pnpm, strict ESLint, Prettier, Husky, lint-staged

02 Market and Differentiation (important)

Research status. Checked on 8 Oct 2026 against official pricing and documentation pages, plus news coverage of two shutdowns. Sources are listed at the end of this section. One statement used in the brainstorm is an assumption, not a fact: that a recruiter gives a portfolio "a minute or two". It shapes priorities but is not used as evidence.

What people use today

Alternative	Example	Built for	Why it fails for Erwin
No-code site builder	Webflow (the current site), Framer	Designers who want to ship visual sites fast	Free plans only give a platform subdomain: Webflow Starter is "Webflow.io domain", and Framer requires a paid plan to connect a domain. A custom domain costs $15 per month (Webflow Basic, billed yearly) or $10 per month (Framer Basic). Shows no React
Starter template	Vercel Portfolio Starter Kit (Next.js, MDX blog, Tailwind v4, OG images, Speed Insights)	Developers who want a blog plus portfolio with little setup	Proves the candidate can deploy a template, not engineer. Generic look, no room for case-study structure or custom motion
Profile platform	GitHub profile README, LinkedIn Featured, Peerlist	Discovery and networking	No space for a case study. Erwin's strongest work is under NDA with no public repo, so these hide it
Hosted profile tool	Read.cv, Bento	One-page profiles	Fragile. Read.cv began winding down in Jan 2025 after Perplexity acquired it. Bento shut down on 13 Feb 2026 and its links now redirect to Linktree

Comparison matrix

Capability	No-code builder	Starter template	Profile platform	This project
React code written by the candidate, open for review	No	Partly (template code)	No	Yes
Case studies for NDA work (architecture, no source)	Yes (current site)	Possible, not included	No	Yes, five
Custom motion system	Yes	No	No	Yes, ported with cleanup
Custom domain without a monthly site fee	No	Yes	No	Yes, later (D-08)
Content edited as data and checked in CI	No	Partly (MDX, no schema check)	No	Yes
Survives the platform changing or closing	No	Yes	No	Yes
Performance and accessibility measured	No	Speed Insights included	Not applicable	Yes, in CI

Positioning

For hiring managers and engineers screening remote React roles, this is a portfolio where **the site itself is the code sample**: a public, tested Next.js repo with a production-grade motion layer and five case studies. Unlike no-code sites, the code is open for review. Unlike starter templates, it is custom work that solves a real lifecycle problem. Unlike profile platforms, it explains NDA work through architecture and decisions, never source.

Three differentiators

#	Differentiator	Objective
1	The site is the code sample: the hardest engineering (motion lifecycle) is visible in the repo and its tests	O2
2	NDA work explained without code: all five case studies at parity, rendered from typed content	O1
3	Owned and measured: free hosting, content as validated data, Core Web Vitals and accessibility checked in CI	O3

What we deliberately do not compete on

A blog or writing, a CMS or visual editing, a redesign or new visuals (parity only, D-05), SEO content marketing, analytics, i18n.

Honest risks to the differentiation

- **Polished motion is cheap now.** Templates and AI site builders produce it quickly, so the difference only shows to someone who opens the repo or asks about it. Mitigation: the footer and README link the repo, and the README has a "How this site works" section with the three proofs.
- **Parity looks identical.** Someone who has seen the Webflow site sees nothing new. That is intended: the evidence is the engineering, not the visuals.
- **A worse React site is negative evidence.** If it is slower or buggier than the Webflow version, it hurts. Mitigation: the link is not sent until the ship checklist passes, and the Webflow site stays as the fallback (D-02).
- **Reviewers may never open the repo.** Mitigation: once measured, the CV and applications name the repo and one proof (for example, the leak test result).

Sources

- Webflow pricing: https://webflow.com/pricing
- Framer pricing: https://www.framer.com/pricing
- Vercel Hobby plan: https://vercel.com/docs/plans/hobby
- Vercel Portfolio Starter Kit: https://vercel.com/templates/next.js/portfolio-starter-kit
- TechCrunch, Perplexity acquires Read.cv (17 Jan 2025): https://techcrunch.com/2025/01/17/perplexity-acquires-read-cv-a-social-media-platform-for-professionals/
- AlternativeTo, Bento to shut down in 2026: https://alternativeto.net/news/2025/12/bento-to-shut-down-in-2026-as-linktree-takes-over-and-offers-migration-path/
- Peerlist: https://www.peerlist.io
- GitHub Docs, managing your profile README: https://docs.github.com/en/enterprise-server@3.15/account-and-profile/how-tos/profile-customization/managing-your-profile-readme

03 Objectives

OBJECTIVE 1

Every page matches the Webflow site, rendered from typed content.

Home, five case studies and a 404 are built from `site.json` and `cases.json` through a validated schema and styled with CSS Modules, and all 28 motion modules from `fx.js` are ported. Copy lives only in JSON; components never hardcode it.

> **Done means: 7 of 7 routes signed off by Erwin against the reference screenshots at 1440, 991, 767 and 390 px (the signed-off screenshots then become Playwright baselines); 28 of 28 `fx.js` modules ported; the content schema test passes in CI.**

OBJECTIVE 2

Motion survives navigation without leaks.

Every effect is scoped with `useGSAP` and reverted on unmount. The page transition owns the cover and reveal steps. Lenis and ScrollTrigger share one animation loop and refresh after fonts load and after each route reveal.

> **Done means: after 10 home to case to back round trips, the live ScrollTrigger count equals the count after the first load, no overlay is left on screen, scroll is unlocked, and the console shows 0 hydration warnings. A Playwright test checks this in CI in Chromium, Firefox and WebKit.**

OBJECTIVE 3

Fast and accessible with motion on.

Content is server-rendered HTML, motion ships only in client islands, and visitors who prefer reduced motion or use touch get a calm, complete site.

> **Done means: Lighthouse mobile performance of 90 or more, with LCP of 2.5 s or less and CLS of 0.1 or less, on `/` and `/work/rover`; 0 serious or critical axe violations on all 7 routes; all content visible with `prefers-reduced-motion: reduce` and with JavaScript disabled. All checked in CI.**

Out of scope

- Blog or MDX (D-04), CMS, i18n, analytics dashboards, light mode, contact form
- Custom domain (later, D-08)
- Redesign, new sections or new content (D-05)
- The Project Artemis case study (already removed from the live site)
- Retiring the Webflow site (optional, D-02)
- Redirects from Webflow paths such as `/work-rover`: the Webflow site keeps its own address, so nothing links there

04 Users and Core Flows

Users

Primary: hiring manager or recruiter Screens candidates for remote React or full-stack roles (Europe first, then the US, Canada and Australia). Arrives from a CV, an application or a LinkedIn message. Wants to learn quickly who Erwin is, his level and proof of shipped work, then get the CV and a way to reach him. Mostly on desktop, sometimes on a phone. The visit fails if the page is slow, breaks on mobile, or animation hides the content.

Secondary: engineer in a technical screen Opens the GitHub repo, reads the README and the motion code, may run it locally. Wants clean architecture, tests and reasons for decisions. Likely question: "Walk me through how your page transition cleans up."

Secondary: Erwin as maintainer Edits copy and adds case studies by changing JSON. Wants CI to tell him when an edit breaks the content.

User stories

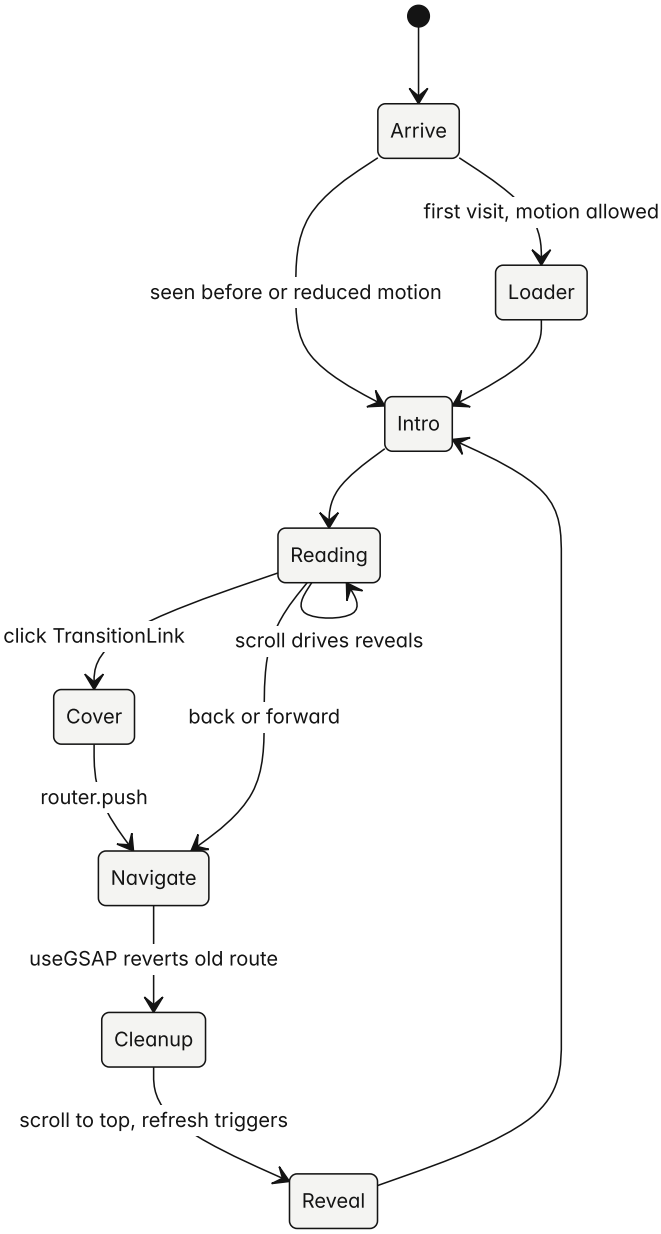
ID	Story	Priority
US-01	As a recruiter, I see name, role, location and availability on the first screen	Must
US-02	As a recruiter, I can open the CV in one click from the nav and from the contact section	Must
US-03	As a recruiter, I can open each of the five case studies from the work section and move on to the next one	Must
US-04	As a recruiter on a phone, I can read every page without horizontal scrolling	Must
US-05	As a visitor who prefers reduced motion, I get all content with no animation	Must
US-06	As a keyboard user, I can reach every link with a visible focus ring; the mobile menu traps focus and closes on Esc	Must
US-07	As a recruiter, I can reach Erwin by email, LinkedIn, GitHub or WhatsApp from the contact section	Must
US-08	As a visitor, a page change plays the wipe with the destination title, and back or forward never leaves a stuck overlay or locked scroll	Must
US-09	As a visitor, the site feels like the Webflow version: smooth scroll, reveals, count-up stats, hero and case intros	Must
US-10	As an engineer, I find a README that explains the architecture and how to run the tests, and CI is green	Must

ID	Story	Priority
US-11	As Erwin, I change copy by editing JSON, and CI fails with a clear message if the edit breaks the schema	Must
US-12	As a visitor, I see the signature effects: first-visit loader, cursor and spotlight, journey pin, rover fleet on motion paths	Should
US-13	As a recruiter sharing the link, the preview shows the right title, description and image	Should
US-14	As a visitor, I can copy the email address and see a confirmation toast	Should
US-15	As a visitor, the polish effects are present: tilt, glare and glow, roll text, magnetic buttons, marquee, scramble labels, read scrub, nav hide and scroll-spy	Could

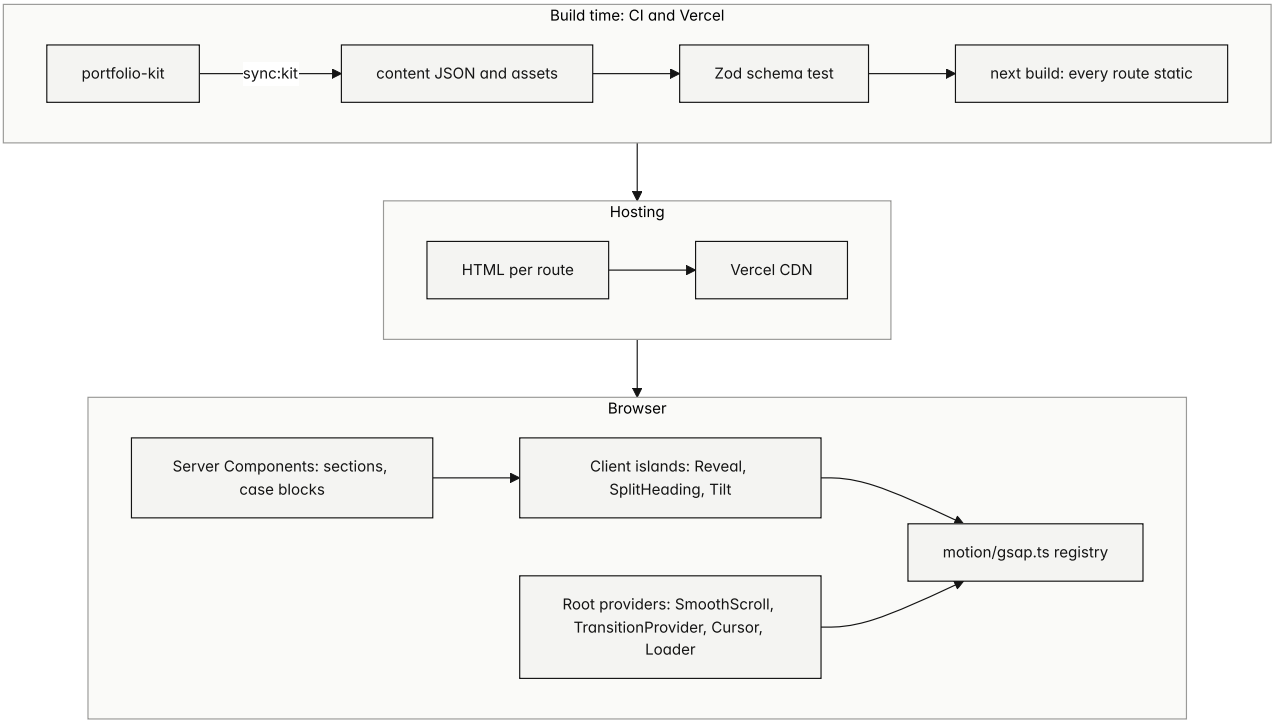
US-15 is "Could" only as build order. D-09 commits to full parity, so it is in scope within the 3-week cap.

Page lifecycle with motion

With reduced motion, Loader, Cover and Reveal are skipped and content is visible at once. Back and forward skip the Cover step but still run Cleanup and Reveal, so the overlay can never stay on screen.


Arrive
first visit, motion allowed
seen before or reduced motion
Loader
Intro
Reading
click TransitionLink
scroll drives reveals
Cover
back or forward
router.push
Navigate
useGSAP reverts old route
Cleanup
scroll to top, refresh triggers
Reveal

05 Architecture



Folder layout (from the build guide): `src/app` (layout, home, `work/[slug]`, not-found), `src/components` (`layout`, `home`, `case`, `ui`, `visuals`), `src/motion` (`gsap.ts`, providers, hooks), `src/content` (JSON, schema, loaders), `public` (images, illustrations, `resume.pdf`).

Server and client rule. A section component never imports GSAP. It renders markup with `data-*` hooks. A thin client wrapper or a hook in a client child adds the motion. If JavaScript fails, the page still reads correctly.

The constraints shaped key decisions

Constraint	So	Rejected alternative
Content is static and already exported as JSON	Every route is generated at build time (`generateStaticParams`, `dynamicParams = false`) (D-10)	Rendering on each request: no dynamic data justifies it
`fx.js` is GSAP, with 28 modules	Keep GSAP, use `@gsap/react` and Lenis (D-11)	Motion (Framer Motion): a rewrite of every effect, with no SplitText or MotionPath equivalent
The App Router has no leave hook, and the wipe shows the destination title	A custom `TransitionProvider` and `TransitionLink` own cover, navigate and reveal (D-12)	React `<ViewTransition>`: CSS-driven, so the dynamic title panel is hard. Kept as the fallback
CSS Modules hash class names	Motion hooks target `data-*` attributes only (D-17, D-18)	Class selectors as in `fx.js`: they stop matching once classes are hashed
Recruiters may arrive on a phone and judge quickly	Server-rendered content, motion in client islands, Lighthouse in CI (D-10, O3)	Whole pages marked `'use client'`: more JavaScript, slower first paint
Formulatrix work is under NDA	The repo holds only what the live site already shows; the kit's `reference/` folder stays local and is gitignored (D-19)	Committing the whole kit
Erwin works on Fedora, with no Safari	Playwright's WebKit engine runs in CI as the Safari stand-in (D-21)	Manual Safari testing

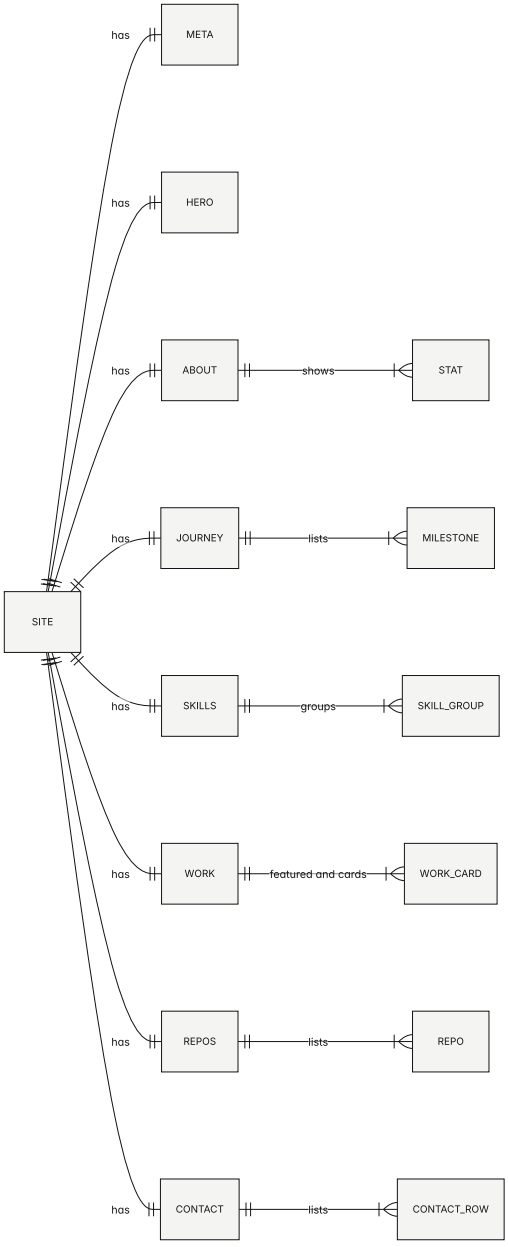
Quality attributes

Attribute	Target	Approach
Performance	Lighthouse mobile 90 or more; LCP 2.5 s or less; CLS 0.1 or less on `/` and `/work/rover`	Static HTML, `next/font`, hero portrait as the priority image, motion in client islands, Lighthouse CI
JavaScript budget	Per-route first-load budget set after the Stage 1 baseline (D-24)	Build output size check in CI
Motion lifecycle	0 leaked ScrollTriggers after 10 round trips; 0 hydration warnings	`useGSAP` scoping and cleanup, refresh after fonts and reveal, a test-only trigger count hook (D-25)
Accessibility	0 serious or critical axe violations on 7 routes; full keyboard access	Reduced-motion gate, `(pointer: fine)` gate for the cursor, focus-visible rings, focus trap in the mobile menu
Resilience	7 of 7 routes readable with JavaScript off	Content in server HTML; motion only adds
Compatibility	Green in Chromium, Firefox and WebKit at 1440, 991, 767 and 390 px	Playwright projects and viewports
Maintainability	Content edits are JSON only; 0 ESLint warnings; strict TypeScript	Zod schema test, strict lint config already in the repo

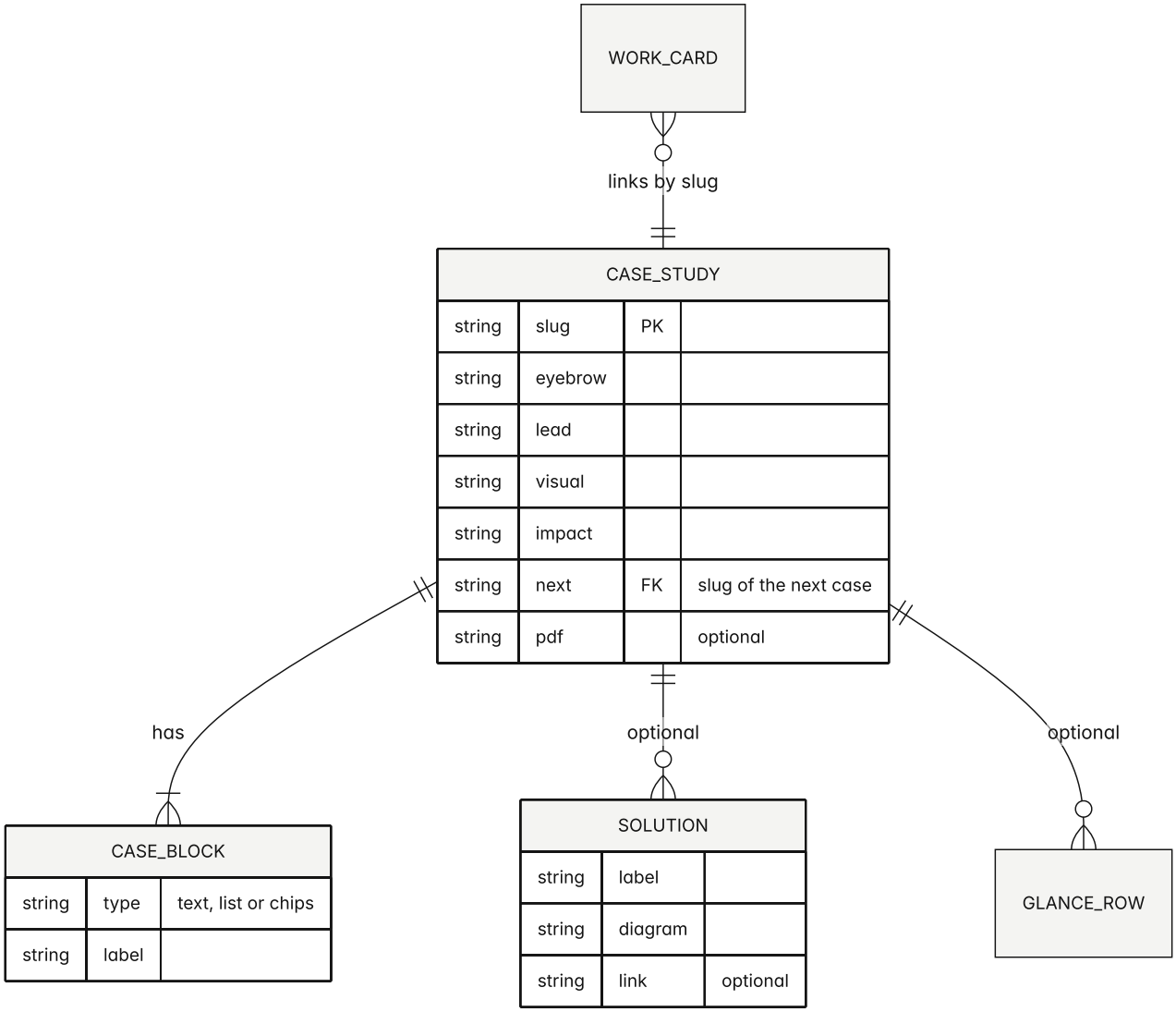
06 Data Model

There is no database. The data model is the content in `src/content/site.json` and `src/content/cases.json`, validated by a Zod schema.

Home content (`site.json`)



Case studies (`cases.json`)



Current content: `site.json` has 10 sections (meta, nav, hero, about, journey, skills, work, repos, contact, footer); `cases.json` has 5 case studies (`rover`, `platform`, `ess`, `promptomat`, `parish`).

Modelling rules

1. JSON is the only source of visible copy. Components never hardcode text.
2. Types come from the Zod schema through `z.infer`. App code imports types only, so Zod never ships to the browser (D-27).
3. `slug` is the ID. Work cards and `next` links reference slugs, and the schema test fails if a slug does not exist.
4. Every asset path (images, illustrations, PDFs) must exist under `public/`. The schema test checks it.
5. Optional case sections (`pdf`, `links`, `ndaNote` or `note`, `stats`, `solutions`, `figure`, `glance`, `reflection`) render only when present.
6. Section ids (`about`, `journey`, `skills`, `work`, `github`, `contact`) are fixed, because nav anchors depend on them.
7. External links open in a new tab with `rel="noopener"`.
8. No em dash in any content string. The schema test rejects them (D-23).

Open questions

- Resolved: the 11 em dashes in `site.json` are replaced during the port (D-23).
- The empty `placeholders` key in `site.json` is a kit leftover. Proposal: drop it from the schema.

07 Design System

Principles

1. **Parity first.** The Webflow design is the spec, and the reference screenshots are the visual target.
2. **Black canvas, one accent.** Lime `#CCFF33` is used only for actions, progress and highlights.
3. **Motion explains, never blocks.** Content is readable before, during and without animation.
4. **Tokens over values.** Components use CSS variables, never raw hex values.
5. **Style and motion stay apart.** CSS Modules style the markup, and `data-*` attributes drive motion (D-18).

Color tokens (WCAG contrast computed)

Token	Value	Used for	Contrast
`--bg--base`	#000000	Page background	Base for the ratios below
`--bg--surface`	#0A0A0A	Cards, loader and wipe panels	
`--bg--raised`	#1A1A1A	Chips, raised cards	
`--border`	#2E2E2E	Dividers, card outlines	1.55:1 on base: decorative only, never the only cue for a control
`--accent`	#CCFF33	CTA fill, progress bar, highlights	17.92:1 as text on base; black text on accent 17.92:1
`--accent--hover`	#D4FF66	Button hover	Black text 18.30:1
`--text--primary`	#EDEDED	Headings and body	17.94:1 on base, 14.87:1 on raised
`--text--secondary`	#D4D4D4	Lead paragraphs	14.17:1 on base, 13.36:1 on surface
`--text--muted`	#A1A1A1	Labels, meta, captions	8.13:1 on base, 7.66:1 on surface, 6.74:1 on raised
`--success` / `--info` / `--error`	#22C55E / #38BDF8 / #F87171	Status dot, rover zones, route conflict	9.22:1, 9.80:1 and 7.59:1 on base

Every text pairing passes WCAG AA, and all but the raised muted pairing pass AAA (7:1).

Type scale

Inter for text, JetBrains Mono for code and labels, both loaded with `next/font` under the variable names the CSS already uses.

Style	Desktop	991 px	767 px	479 px	Weight, line height
`heading-xl` (hero)	76	60	46	39	700, 1.06
`heading-case`	72	56	40	34	700, 1.06
`heading-contact`	68	54	40	34	700, 1.08

Style	Desktop	991 px	767 px	479 px	Weight, line height
`heading-l` (sections)	48	40	32	32	600, 1.15
`stat-number`	44				700, 1
`case-lead`	20				1.65
`case-text`	19				1.7
Buttons	16				600, 1.2
`nav-link`	15				500
`chip`	14				500, 1.3
`eyebrow` / `case-label`	13 / 12				500, letter spacing 0.14em
Body base	14				line height 20 px

Values are in px, taken from `styles/site.css`. Empty cells keep the desktop value.

Spacing, radius, motion

- **Layout:** container max width 1248 px with 24 px side padding; sections have 128 px top padding.
- **Gaps:** mostly 24, 20, 48, 32 and 16 px; card padding 28 to 32 px.
- **Radius:** 999 px for pills and buttons, 18 to 24 px for cards, 10 to 14 px for small elements, 50% for dots and avatars.
- **Breakpoints:** 991, 767 and 479 px, as `max-width` queries in that order.
- **Motion:**
  - CSS transitions run 200 to 300 ms on color, border and transform, using `--fx-ease: cubic-bezier(.16, 1, .3, 1)`.
  - GSAP eases: `expo.out`, `expo.inOut`, `power3.out`, `power4.out`, `back.out(3)`.
  - Reduced motion turns every effect off.

Component inventory and states

Component	States
Button (primary, secondary, nav)	Default, hover (accent hover fill or border and text change), focus-visible ring, pressed; week 2: magnetic pull (fine pointer only), roll text on hover
NavLink	Default, hover, current section (scroll-spy underline), focus-visible
Nav	Visible at the top; hidden on scroll down and shown on scroll up (week 2); mobile menu closed or open (focus trapped, Esc closes, page scroll locked)
Chip	Default, primary (accent), revealed (staggered)
StatusPill	Default with a live success dot
StatCard	Start value, counting, final value; reduced motion shows the final value
WorkFeatured, WorkCard	Default, hover (tilt, glare, glow with a fine pointer), focus-visible, visual loop playing or paused off-screen

Component	States
RepoCard	Default, hover (glow, arrow nudge), focus-visible
ContactCard, CopyEmail	Idle, copied (toast shown), copy failed (falls back to a `mailto:` link)
Toast	Hidden, shown, hiding
Cursor	Off (touch, coarse pointer or reduced motion), default disc, hug (links and buttons), I-beam (text), card (dot and lift)
Loader	Skipped (seen this session, or reduced motion), counting, exiting
Wipe	Idle, covering (lime panel, black panel, destination title), revealing
ScrollProgress	0 to 100% scrubbed; hidden with reduced motion
Journey	Desktop pinned with scrubbed milestones; mobile vertical progress; reduced motion as a static list
CaseBlock list	Hidden, then items pop in; reduced motion shows them at once
RoverFleet, mini visuals	Moving on paths, tooltip on hover or focus, paused off-screen; reduced motion static
Images	Space reserved by aspect ratio (no layout shift), loaded

08 Decision Log

Rule: a decision changes in this log first, then in code.

ID	Decision	Alternatives	Why	Status
D-01	Name the GitHub repo and package `software-engineer-portfolio`	Keep `portofolio_dev`; `software_engineer_portofolio`	Credibility: correct spelling, and hyphens are the GitHub convention. The local folder name does not matter	**Accepted**
D-02	The Webflow site stays live in parallel. Erwin sends the React link when a role wants React evidence; retiring Webflow is optional	Retire Webflow at launch	Zero-risk fallback while the React version matures	**Accepted**
D-03	Ship v1 in 1 week	2 weeks	First timebox	**Superseded by D-09**
D-04	No blog in v1	MDX blog, as in the starter kit	Does not fit the timebox, and an empty blog looks worse than none	**Accepted**
D-05	v1 is a faithful copy of the Webflow site, with no new features	Redesign	Content and design are already validated; the evidence is the engineering	**Accepted**
D-06	Mixed mode: Erwin writes components and pages; Claude Code mentors and does the chores	Claude Code writes everything; Erwin writes everything	Only code Erwin writes and can explain counts as evidence, and chores would eat the timebox	**Accepted**
D-07	Help ladder (explain, hint, skeleton, pair) with a 30-minute rule; every spec opens with a Concepts box	Ad-hoc help	Getting stuck on new APIs is the top schedule risk	**Accepted**
D-08	Custom domain later	Buy one now	Not needed to prove the work; `vercel.app` is fine for v1	**Accepted**
D-09	Full parity, targeted at 2 weeks with a 3-week hard cap. Week 1 ends shippable (v1 scope on `vercel.app`)	1 week with effects cut	Erwin prefers extending the time to cutting effects	**Accepted**
D-10	Next 16 App Router, every route static	Rendering per request	No dynamic data; fastest first paint, free hosting	**Accepted**

ID	Decision	Alternatives	Why	Status
D-11	GSAP, `@gsap/react` and Lenis	Motion (Framer Motion)	`fx.js` is GSAP; switching means rewriting all 28 modules	Accepted
D-12	Custom `TransitionProvider` and `TransitionLink`, with `<ViewTransition>` as the fallback	`<ViewTransition>` first	The wipe shows the destination title; owning leave and enter is the hard problem to prove	Accepted
D-13	Content as JSON, typed, plus a schema check in tests	MDX, a CMS	Content is already exported; a broken edit should fail CI, not production	Accepted
D-14	Playwright for end-to-end checks (smoke, back and forward, leak count, axe, reduced motion); Vitest for the content schema	Component unit tests	The proof points are browser behaviour	Accepted
D-15	GitHub Actions on every PR (lint, typecheck, Vitest, build, Playwright, Lighthouse) and a Vercel preview per PR	Manual checks	Every acceptance check must pass in CI	Accepted
D-16	Keep pnpm, strict ESLint, Prettier, Husky and lint-staged	None	Already set up in the repo	Accepted
D-17	CSS Modules per component	Global `site.css`; Tailwind	Shows React structure; Tailwind would mean rewriting 2,273 lines of CSS	Accepted
D-18	Motion hooks target `data-*` attributes only	Class selectors	CSS Modules hash class names; keeps style and motion independent	Accepted
D-19	The repo holds only what the live site already shows; the kit's `reference/` folder stays local and is gitignored	Commit the whole kit	NDA safety and repo size	Accepted
D-20	Claude Code reads the docs bundled in `node_modules/next/dist/docs` before writing Next-specific code	Rely on training data	Next 16 changed APIs; the repo's AGENTS.md already asks for this	Accepted
D-21	Playwright's WebKit engine runs in CI as the Safari stand-in	Manual Safari checks	Erwin works on Fedora, with no Safari	Accepted
D-22	After v1, both sites stay full showcases: every content change goes to the JSON and to the Webflow site in the same session, and `progress.md` records both	JSON first with Webflow critical fixes only; Webflow leads; freeze Webflow	Webflow stays the showcase for Webflow roles and React for React roles (D-02), so neither may go stale	Accepted

ID	Decision	Alternatives	Why	Status
D-23	Replace all 11 em dashes in `site.json` during the port, and add a schema rule that rejects them	Keep exact parity	Erwin's writing rule; a one-time content fix	**Accepted**
D-24	Set a per-route first-load JavaScript budget from the Stage 1 baseline, and fail CI above it	A fixed number now	No baseline exists yet, so a number now would be invented	**Accepted**
D-25	Expose the live ScrollTrigger count through a test-only window hook, enabled only in the end-to-end build	Count DOM side effects instead	The leak test needs a direct, reliable number	**Accepted**
D-26	If 3 weeks is not enough, re-plan with Erwin before cutting anything	Cut list agreed now	Erwin chose time over cuts	**Accepted**
D-27	Zod as the schema library, with types inferred through `z.infer` and imported as types only	Hand-written TS interfaces plus a custom check	One source for types and validation, at no runtime cost	**Accepted**

09 Delivery Plan

Days count from the first working day after sign-off. Week 1 is Days 1 to 7, week 2 is Days 8 to 14, and week 3 (Days 15 to 21) is buffer only: no new scope enters it.

1

Foundation (Days 1 to 2)

- **Scope:** F-01 to F-03.
  - Repo rename and remote, kit sync, content schema and loaders.
  - CI pipeline and Vercel previews.
  - Tokens, fonts, base reset and shared UI.
  - Layout shell (Nav, Footer, Container).
- **Acceptance check (CI):**
  - The schema test passes, and fails on a broken fixture.
  - Smoke test: `/` returns 200 with no console errors in three engines.
  - Nav anchors exist, and below 991 px the mobile menu opens, traps focus and closes on Esc.
  - A Vercel preview is posted to the PR.

2

Static parity (Days 3 to 5)

- **Scope:** F-04 to F-08.
  - Home (7 sections), the case template for 5 slugs, and the 404.
  - Metadata, Open Graph image, sitemap and robots.
- **Acceptance check (CI):**
  - All 7 routes load in three engines with no console errors.
  - The `next` links loop through all five cases back to Rover.
  - 0 serious or critical axe violations.
  - No horizontal overflow at 390 px, and all content visible with JavaScript disabled.
  - Erwin signs off each route against the reference screenshots, and the signed-off screenshots (taken with reduced motion) become Playwright baselines.

3 Motion foundation and navigation (Days 6 to 7), the end of week 1

- **Scope:** F-09 and F-10.
  - GSAP registry and SmoothScroll on one loop, the reduced-motion gate.
  - Reveal, count-up, progress bar.
  - Page transition, hero and case intros.
  - Leak test and Lighthouse CI.
- **Acceptance check (CI):**
  - Leak test: 10 round trips with a constant trigger count, no stuck overlay, scroll unlocked and at the top after navigating.
  - 0 hydration warnings.
  - Reduced-motion test: everything visible, nothing transformed.
  - Lighthouse mobile 90 or more on `/` and `/work/rover`.
  - **Production deploy on `vercel.app` (v1).**

4 Signature effects (Days 8 to 10)

- **Scope:** F-11 to F-15.
  - Loader.
  - Cursor and spotlight.
  - Split headings, scramble labels and read scrub.
  - Journey pin.
  - Rover fleet and minis.
- **Acceptance check (CI):**
  - The loader shows once per tab session and never with reduced motion.
  - No cursor element under touch emulation or reduced motion.
  - The journey pin causes no layout shift (CLS stays at 0.1 or less).
  - Rover tooltips are reachable by keyboard.
  - The leak test, axe and Lighthouse stay green.

5

Polish and ship (Days 11 to 14)

- **Scope:** F-16 to F-19.
  - Card interactions and roll text, magnetic buttons, marquee, hero scroll-out and case parallax.
  - Nav hide, scroll-spy, copy email and toast.
  - README "How this site works", ship checklist, production deploy.
- **Acceptance check (CI):**
  - All suites green in three engines.
  - The copy-email test shows the toast.
  - The scroll-spy test marks the current section.
  - 28 of 28 `fx.js` modules ticked in `progress.md` .
  - Ship checklist complete.

Metrics

Metric	Target	How measured	Actual
Routes at visual parity	7 of 7	Erwin's sign-off, then Playwright screenshot baselines	
`fx.js` modules ported	28 of 28	Checklist in `progress.md`	
ScrollTriggers leaked after 10 round trips	0	Playwright leak test (D-25)	
Hydration warnings	0	Playwright console listener on all routes	
Lighthouse mobile performance	90 or more on `/` and `/work/rover`	Lighthouse CI	
LCP / CLS (lab, mobile)	2.5 s or less / 0.1 or less	Lighthouse CI	
Serious or critical axe violations	0 on 7 routes	`@axe-core/playwright`	
Routes readable with JavaScript off	7 of 7	Playwright with JavaScript disabled	
Browser engines green in CI	3 of 3	Playwright projects	
First-load JavaScript per route	Budget from Stage 1 (D-24)	Build output check in CI	
Calendar time to full parity	14 days or less (cap 21)	Dates in `progress.md`	
Help-ladder pair sessions (rung 4)	Tracked, no target	Log in `progress.md`	

10 Feature Development Workflow

The feature loop



1. **Pick** the next feature: `progress.md` names it.
2. **Write the spec.** Claude Code drafts it from the template below, including the Concepts box.
3. **Erwin reviews** and approves the spec. No code before approval.
4. **Tests first.** Claude Code writes the failing Playwright or Vitest tests for the acceptance criteria (a chore).
5. **Implement in small commits.** Erwin writes the components and pages, and Claude Code mentors (help ladder, D-07). Claude Code implements chore tasks.
6. **Verify:** tests green locally, compared with the reference screenshot at the four widths.
7. **PR and checklist.** The PR description ends with Erwin's "How it works" teach-back in his own words. Claude Code reviews and asks 2 questions.
8. **Update docs:** `progress.md` (what changed, weak spots, next step) and `decisions.md` if anything changed.

Rules

- One feature per branch and PR: `feat/F-04-hero-about`, `chore/F-01-foundation`.
- No code before an approved spec.
- Conventional commits, enforced by a commit-msg hook.
- Claude Code never writes component or page code unless Erwin asks for rung 4, and every rung 4 assist is logged in `progress.md`.
- Stuck for 30 minutes on one thing means moving up one rung of the help ladder.
- No em dash in code comments, docs, commit messages or PR text.

Feature backlog

ID	Feature	Stories	Stage	Depends on	Size	`fx.js` modules
F-01	Repo foundation: rename and remote, kit sync, content schema and loaders, CI, Vercel	US-10, US-11	1	None	M	
F-02	Tokens, fonts, base reset, CSS Modules conventions, shared UI (Button, Chip, Eyebrow, SectionHead, StatusPill, Card)	US-04, US-06	1	F-01	M	
F-03	Layout shell: Nav with mobile menu, Footer, Container	US-02, US-06	1	F-02	M	
F-04	Home: Hero, About and StatCard	US-01	2	F-03	M	
F-05	Home: Journey and Skills	US-01	2	F-03	M	

ID	Feature	Stories	Stage	Depends on	Size	`fx.js` modules
F-06	Home: Work, Repos and Contact	US-03, US-07	2	F-03	M	
F-07	Case template `/work/[slug]`, all optional sections, 404	US-03, US-04	2	F-03	L	
F-08	Metadata, Open Graph image, sitemap, robots	US-13	2	F-07	S	
F-09	Motion foundation: registry, SmoothScroll, reduced-motion gate, Reveal, count-up, progress bar, Lighthouse CI	US-05, US-09	3	F-04 to F-07	M	lenis, progress, reveals
F-10	Page transition, hero and case intros, leak test	US-08, US-09	3	F-09	L	links, enter, heroIntro, caseIntro
F-11	First-visit loader with no flash	US-12	4	F-10	S	loader
F-12	Cursor and spotlight	US-12	4	F-09	M	cursor, spotlight
F-13	Split headings, scramble labels, read scrub	US-15	4	F-09	M	split, scramble, read
F-14	Journey pin and mobile progress	US-12	4	F-09	M	journey
F-15	Rover fleet motion paths and tooltips, mini visuals	US-12	4	F-09	M	rover, minis
F-16	Card tilt, glare and glow, roll text, magnetic buttons, "move." jiggle	US-15	5	F-09	M	interactions, roll, magnetic, magneticBase, move
F-17	Marquee, hero scroll-out, work visual scale, case parallax and next-title fill	US-15	5	F-09	M	marquee, heroOut, work, case
F-18	Nav hide, scroll-spy, copy email and toast	US-14, US-15	5	F-03, F-09	S	nav, spy, copy
F-19	Ship: README "How this site works", ship checklist, production deploy	US-10	5	All	S	

The `fx.js` column covers all 28 modules exactly once.

Feature spec template

Every spec in `docs/features/F-xx-name.md` has a Concepts box followed by ten parts:

0. **Concepts box:** what is new in this feature, a docs link for each concept, 2 to 3 lines on why it matters.
1. **Goal:** one paragraph.
2. **Acceptance criteria:** numbered, written as Given / When / Then.
3. **API contract:** component props, hook signatures and the content types read. This is a static site, so this replaces a REST contract.
4. **Data changes:** JSON, schema or asset changes.
5. **Frontend states:** every visual and motion state, including reduced motion.
6. **Edge cases.**
7. **Test plan:** each test mapped to an acceptance criterion.

8. **Task breakdown:** one commit per line, with an owner (Erwin or Claude Code).
9. **Out of scope.**
10. **Decisions referenced.**

Worked example: F-01 Repo foundation

0. Concepts box

- **Zod and `z.infer`:** one schema gives both runtime validation and TypeScript types. https://zod.dev
- **Playwright projects:** one config runs the same tests in Chromium, Firefox and WebKit. https://playwright.dev/docs/test-projects
- **GitHub Actions:** a workflow file runs the checks on every pull request. https://docs.github.com/actions
- **Vercel Git integration:** every PR gets its own preview URL. https://vercel.com/docs/git
- **Next 16 file conventions:** read the docs bundled in `node_modules/next/dist/docs` before touching `app/` (D-20).

1. Goal. A renamed, connected repository where the current kit content and assets live inside the app, typed and validated. Every PR runs the full check pipeline and gets a Vercel preview. No visual work yet.

2. Acceptance criteria

1. Given the GitHub account Blackpossum, when the repo list is opened, then `software-engineer-portfolio` exists with `main` pushed and the package name matches.
2. Given the kit at `~/Downloads/portfolio-kit`, when `pnpm sync:kit` runs, then `public/images`, `public/illustrations`, `public/resume.pdf` and `src/content/*.json` match the kit, the four stale Artemis-era files are removed, and nothing from `reference/` is copied.
3. Given valid content, when `pnpm test:unit` runs, then the schema test passes. Given a fixture with an unknown `next` slug, a missing asset file or an em dash, then it fails with a message naming the field.
4. Given a pull request, when CI runs, then lint, typecheck, unit tests, build and the Playwright smoke test (three engines) all run and are required to merge.
5. Given the smoke test, when it loads `/`, then the response is 200 and the console has no errors.
6. Given a pull request, when Vercel builds it, then a preview URL appears on the PR.
7. Given the repository, when CI lists tracked files, then no path under `reference/` is tracked.

3. API contract

- `src/content/schema.ts` exports the `SiteSchema` and `CaseStudySchema` Zod schemas, and the `Site`, `CaseStudy` and `CaseBlock` types.
- `src/content/index.ts` exports:
  - `getSite(): Site`
  - `getCases(): CaseStudy[]`
  - `getCase(slug: string): CaseStudy | undefined`
- `package.json` scripts: `sync:kit`, `typecheck`, `test:unit`, `test:e2e`, `lint`, `build`.

4. Data changes

- Add `src/content/site.json` and `src/content/cases.json` from the kit.
- Add `public/resume.pdf`.

- Replace `public/illustrations` (add 23 files; remove `card-arch.svg`, `card-code.svg`, `card-streak.svg` and `case-artemis.svg`).
- Refresh `public/images`.
- Bring `src/styles/tokens.css` and `src/styles/site.css` up to the current kit. `site.css` stays as a porting reference only and is not imported.

5. Frontend states. Only a placeholder home page that replaces the create-next-app page and its leftover Tailwind classes.

6. Edge cases

- Kit path missing: the sync script exits with a clear message and changes nothing.
- Paths are case-sensitive on Vercel's Linux builders.
- Em dashes: the 11 in `site.json` are replaced in F-01 and the schema rejects new ones (D-23). Erwin reviews the new wording, since it is his copy.
- The `placeholders` key is dropped from the schema.
- `sharp` builds are disabled in `pnpm-workspace.yaml`, so check that `next/image` still works on Vercel.

7. Test plan

Test	Criteria
`content.test.ts`: valid content passes; mutated fixtures (bad slug, missing asset, em dash) fail	AC3
`smoke.spec.ts`: `/` returns 200 with no console errors, in three engines	AC4, AC5
CI step that fails if `git ls-files` lists `reference/`	AC7
Sync script dry run, then a diff against the kit	AC2
Manual checks: repo, package name, Vercel preview on the first PR	AC1, AC6

8. Task breakdown (one commit each)

1. `chore: rename package to software-engineer-portfolio` (Claude Code). Erwin creates the GitHub repo and adds the remote.
2. `chore: replace default page with placeholder` (Claude Code)
3. `chore: add kit sync script and sync content, assets and resume` (Claude Code)
4. `test: add failing content schema test` (Claude Code)
5. `feat: add zod content schema and typed loaders` (Claude Code)
6. `fix(content): replace em dashes in site.json` (Claude Code; Erwin approves the wording)
7. `test: add Playwright with three browser projects and smoke test` (Claude Code)
8. `ci: add GitHub Actions workflow for lint, typecheck, unit, build and e2e` (Claude Code)
9. `chore: add commitlint commit-msg hook` (Claude Code)
10. `docs: README with run and test commands` (Claude Code)
11. Connect the Vercel project in the dashboard (Erwin; no commit)

9. Out of scope. Any visual component, tokens and fonts in use (F-02), Lighthouse CI (F-09), custom domain.

10. Decisions referenced. D-01, D-06, D-13, D-14, D-15, D-16, D-19, D-20, D-23, D-27.

Definition of Done

- [ ] Every acceptance criterion passes in CI (or is checked manually where the spec says so)
- [ ] Lint, typecheck and build pass with 0 warnings
- [ ] UI features: matches the reference screenshot at 1440, 991, 767 and 390 px; keyboard reachable; works with reduced motion
- [ ] No console errors or hydration warnings
- [ ] Erwin can explain every line he wrote; rung 4 assists are logged
- [ ] The PR has the teach-back "How it works" section
- [ ] progress.md updated, and decisions.md too if anything changed
- [ ] Content changed? The Webflow site got the same change in this session (D-22)
- [ ] Conventional commits, and no em dash anywhere

11 Risks and Expected Change

Risk	Impact	Mitigation
Erwin gets stuck on new APIs (App Router, `useGSAP`, Lenis)	High: a lost day is 7% of a 2-week target	Help ladder, 30-minute rule, a Concepts box in every spec, working snippets in the build guide (D-07)
The timebox slips because of applications and interviews	High	Week 1 ends shippable; `progress.md` is updated every session; week 3 is buffer only; re-plan before cutting (D-09, D-26)
Next 16 differs from what Claude was trained on	Medium	Read the bundled docs first (D-20)
Hydration errors from SplitText and roll text	Medium	DOM-rewriting effects only inside `useGSAP`, reverted in cleanup, never on the server; Playwright fails on hydration warnings
ScrollTrigger leaks or wrong positions after navigating	High: this is proof point O2	`useGSAP` cleanup, refresh after fonts and each route reveal, leak test in CI (D-25)
Stale assets: the repo has the 3 Oct kit	Medium	F-01 syncs from the current kit
NDA: the public repo leaks more than the site shows	High	The repo holds only what the live site shows; `reference/` stays local (D-19)
Content drift between Webflow and React	Medium: two live sites	Every content change updates both in the same session, and the Definition of Done checks it (D-22)
Heavy motion hurts Core Web Vitals	Medium	Client islands, JavaScript budget (D-24), Lighthouse CI
Safari-only bugs (`background-clip: text`, `@property`)	Medium	WebKit in CI (D-21), `@supports` guards from `motion.css`
Vercel Hobby is non-commercial	Low	Fine for a personal portfolio; move to Pro or another host if the site ever sells services

Before and after at a glance (planned)

	Before (Webflow)	After (React, planned)
Public code	None	Public repo with tests and CI
Address	`webflow.io` subdomain; a custom domain needs a paid plan	`vercel.app` now, own domain later; hosting free
Content edits	Designer session and publish	JSON edit and PR, validated in CI
Automated tests	0	Content schema, end-to-end in 3 engines, axe, Lighthouse
Navigation	Full page load on every link	Client navigation with a wipe, measured to be leak-free
Accessibility checks	Manual	axe in CI on 7 routes
Performance	Not measured	Lighthouse mobile 90 or more, measured in CI

12 Handoff to Claude Code

Repo files

File	Purpose
`docs/design.md` and `docs/design.pdf`	This document, the source of truth
`docs/decisions.md`	The decision log only; decisions change here first
`docs/brainstorm-notes.md`	Raw brainstorm summary, including rejected ideas and why
`docs/features/F-01-repo-foundation.md`	The worked spec; later specs land in this folder
`docs/progress.md`	Session log. Starts with "next feature = F-01" and the 28-module checklist
`CLAUDE.md`	Order of authority, session ritual, mentor and implementer roles, help ladder, commands, writing rules. It keeps the `@AGENTS.md` import that Next manages

Order of authority

1. Erwin's current message
2. `docs/decisions.md`, Accepted rows
3. `docs/design.md`
4. Existing code conventions
5. Claude Code's own judgement, only for details nothing above covers, and logged in `progress.md`

Session ritual

1. Read `CLAUDE.md`, `docs/design.md`, `docs/decisions.md` and `docs/progress.md` before any code.
2. Restate in 5 lines: the current stage, its acceptance check, the feature in progress, the decisions that apply, and Erwin's weak spots from the last session. Wait for Erwin to confirm.
3. Open with the feature's Concepts box (mentor brief).
4. Work on one feature per session, following the feature loop. Help ladder on request; the 30-minute rule applies.
5. If something is not in the docs, ask; do not decide.
6. End by appending to `progress.md`: what changed, the teach-back result and 2 questions, weak spots, new decision rows, next step.

Design doc to case study mapping (for Phase 6)

Design doc section	Case study section
1. The Problem	The Challenge
2. Market and Differentiation	Why it matters
3. Objectives	Objectives, with real outcomes
9. Delivery Plan (stages)	Solution and outcome by stage
8. Decision Log (best 3 rows)	The constraints that shaped key decisions
9. Metrics (Actual column)	Measured results
11. Before and after	Before and after at a glance, measured
11. Risks, plus `progress.md` notes	What made this hard